

Aydin Javadov

PhD Candidate at ETH Zurich

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EDUCATION

- **ETH Zurich** Zurich, Switzerland
PhD Candidate in Machine Learning, Associated Researcher at ETH AI Center *Nov. 2024 – Present*
Focusing on:
Human-AI Alignment, Explainable AI, Multimodality, Large Language Models
Supervised by:
Prof. Dr. Florian von Wangenheim (Mobiliar Lab for Analytics @ ETH Zurich) &
Prof. Dr. Bjoern Schuller (GLAM @ Imperial College London & CHI @ TUM)
- **Technical University of Munich** Munich, Germany
M.Sc. in Data Engineering & Analytics; Graduation with Distinction (in Top $\leq 10\%$) *April 2021 – July 2024*
German Grade: 1.5

Thesis (done during research stay at BMW Research):
Explainable AI for Graph Representation Learning and Clustering Algorithms (German Grade: 1.0)
Supervised by:
Prof. Dr. Viktor Leis
- **Korean Advanced Institute of Science and Technology** Daejeon, South Korea
(Exchange Stay) Computer Science *February 2018 – June 2018*
- **ADA University** Baku, Azerbaijan
B.Sc. in Computer Engineering; Graduation with Distinction (in Top $\leq 1\%$) *September 2016 – June 2020*
US Grade: 3.89 (German Equivalent: 1.1)

Thesis:
Analysis of Distributed Temperature Sensor Data Using Machine Learning
Supervised by:
Prof. Dr. Samir Rustamov

EXPERIENCE

- **ETH Zurich** Zurich, Switzerland
PhD Candidate in Machine Learning & ETH AI Center Associated Researcher *Nov. 2024 – Present*
 - Supervised by Prof. Dr. Florian von Wangenheim (Mobiliar Lab for Analytics @ ETH Zurich) and Prof. Dr. Bjoern Schuller (GLAM @ Imperial College London & CHI @ TUM).
 - Focusing on: Human-AI Alignment, Explainable AI, Multimodality, Large Language Models.
- **BMW Group** Munich, Germany
Machine Learning Research Student *April 2023 – May 2024*
 - Led the project on Explainable AI integration to the graph learning for engine-problems – pushed to the production pipeline.
 - Focusing on: Large Language Models, Time Series Analysis, Graph Representation Learning, Explainable AI.
- **BMW Group** Munich, Germany
Master's Thesis Research *October 2022 – April 2023*
 - Thesis title: *Explainable AI for Graph Representation Learning and Clustering Algorithms*.
 - Technologies: Python, PyTorch, AWS, Git.
 - Graded: 1.0 (German System).
- **Technical University of Munich** Munich, Germany
Guided Research at the Chair of Computational Imaging and Inverse Problems *April 2022 – November 2022*
 - Project title: *Explainable AI and Computer Vision for Clinical Decision Support in Dermatology*.
 - The subject was understanding and implementation of several interpretability techniques for deep learning models for skin lesion classification.

- The work was on the theme of human-centered explainable AI and involved close collaboration with Munich University Clinic physicians.

- **novuter GmbH**

Munich, Germany

- *Data Science Working Student*

September 2021 – May 2022

- Extracted raw data and transformed the data stories on different domains (e.g. finance) to optimize the decision making.
- Technologies: PostgreSQL, JavaScript, Python, R.

- **ATL Tech – AI Lab**

Baku, Azerbaijan

- *Artificial Intelligence Intern*

October 2019 – February 2020

- Took part in the advanced research of Speech Recognition in Dialog Systems for the Azerbaijani Language.
- Technologies: Python, pandas, numpy.

SELECTED PUBLICATIONS

- **ICLR 2026:** Y. Wang, Y. Hou, **A. Javadov**, M. Akhtar, M. Sachan. “Compose and Fuse: Revisiting the Foundational Bottlenecks in Multimodal Reasoning”.
- **NeurIPS 2025 – Learning from Time Series for Health Workshop:** **A. Javadov**, S. Garibov, T. Hoesli, Q. Sun, F. von Wangenheim, J. Ollier, B. Schuller. “RAXSS: Retrieval-Augmented Sparse Sampling for Explainable Variable-Length Medical Time Series Classification”.
- **CHI 2025 – Envisioning the Future of Interactive Health:** **A. Javadov***, M. Baumgartner*, R. Ney, F. von Wangenheim, J. Ollier. “BioSyncHRI: Synchronizing Human Robot Interaction via Real-Time Biosignal Adaptation”.

AWARDS AND PARTICIPATIONS

- **84th out of 78,545 Students (top 0.001%):** National University Entrance Exam of Azerbaijan – July 2016.
- **Winner:** Hackathon HackaTUM, Technical University of Munich & Carl Zeiss AG (ZEISS) – November 2021.
- **228th out of 40,875 Students (top 0.005%):** Primary School Republic Examination of Türkiye – November 2010.
- **Global Korea Scholarship:** Ministry of Education of the Republic of Korea – February to June 2018.
- **2x Rector’s List of Honour & 3x Dean’s List of Honour:** Merit-Based Scholarship at ADA University – 2017–2020.
- **Organizer of ‘Purple Comet’ International Math Olympiad:** ADA University – April 2019.

TECHNICAL SKILLS

- **ML/AI Concepts:** Representation Learning, LLM, XAI, Deep Learning, Computer Vision, Machine Learning, Time Series, Deep Generative Models, Uncertainty Quantification.
- **Application Areas:** Medicine, Cognitive Modelling, Human Factors, Finance, Signal Processing, Dynamic Systems.
- **Other:** Python, Git, Bash, PyTorch, Azure, AWS, Spark.

LANGUAGES

- **Native:** Turkish, Russian, Azerbaijani.
- **Fluent:** English.
- **Elementary:** German.
- **Beginner:** Korean.

REFEREES

- **Academic:** Prof. Dr. Bjoern Schuller (Imperial College London & TUM); Prof. Dr. Viktor Leis (TUM); Prof. Dr. Felix Dietrich (TUM); Prof. Dr. Florian von Wangenheim (ETH Zurich).
- **Industry:** Prof. Dr. Tobias Lasser (NVIDIA); Dr. Raphael Weingartner (BMW); Dr. Iryna Bastian (BMW).